

SECR2043 OPERATING SYSTEMS

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 Section : 03 _____

Marks

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write **"No output"** instead.

1.	Display your current working directory.	
	Command	Output
	pwd	C:\Users\jingyi>ssh chohjingyi@192.168.56.10 chohjingyi@192.168.56.10's password: Last login: Thu Apr 17 04:33:12 2025 from 192.168.56.1 chohjingyi@secr2043:~\$ pwd /home/chohjingyi
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
	ls -a ~	chohjingyi@secr2043:~\$ ls -a ~bash_logout .bashrc .bash_history .cache .lessshst .local .profile .sudo_as_admin_successful file2.txt filelist.txt 'nano myfile.txt' practice.txt sample.txt workspace
3.	Create a new directory named "os_lab" in your home directory.	
	Command	Output
	mkdir ~/os_lab	No output
4.	Change your current working directory to "os_lab".	
	Command	Output
	cd ~/os_lab	No output
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it.	
	Command	Output
	echo "Hello, world!" > hello.txt	No output

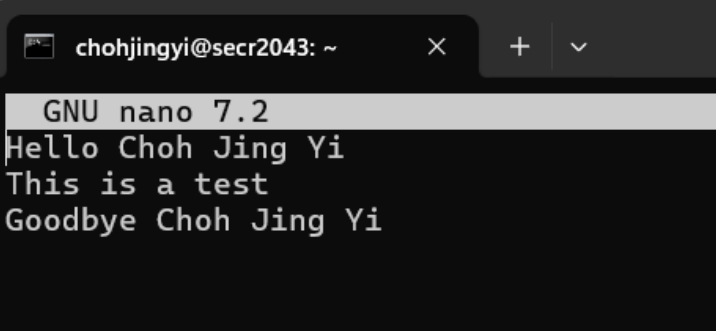
6.	Display the content of "hello.txt".	
	Command	Output
	cat hello.txt	chohjingyi@secr2043:~\$ cat hello.txt Hello, world!
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory.	
	Command	Output
	cp hello.txt hello_copy.txt	No output
8.	Rename "hello_copy.txt" to "hello_bak.txt".	
	Command	Output
	mv hello_copy.txt hello_bak.txt	No output
9.	Delete "hello.txt".	
	Command	Output
	rm hello.txt	No output
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	Output
	ls -lR ~	chohjingyi@secr2043:~\$ ls -lR ~ /home/chohjingyi: total 40 -rw-r--r-- 1 chohjingyi chohjingyi 82 Apr 16 15:49 combined.txt -rw-r--r-- 1 chohjingyi chohjingyi 41 Apr 16 15:42 file2.txt -rw-r--r-- 1 chohjingyi chohjingyi 61 Apr 16 15:45 filelist.txt -rw-r--r-- 1 chohjingyi chohjingyi 14 Apr 17 04:49 hello.txt -rw-r--r-- 1 chohjingyi chohjingyi 14 Apr 17 04:50 hello_bak.txt -rw-r--r-- 1 chohjingyi chohjingyi 12 Apr 17 03:28 'nano myfile.txt' drwxr-xr-x 2 chohjingyi chohjingyi 4096 Apr 17 04:48 os_lab -rw-r--r-- 1 chohjingyi chohjingyi 758 Apr 17 03:40 practice.txt -rw-r--r-- 1 chohjingyi chohjingyi 725 Apr 17 03:39 sample.txt drwxr-xr-x 3 chohjingyi chohjingyi 4096 Apr 17 04:37 workspace /home/chohjingyi/os_lab: total 0 /home/chohjingyi/workspace: total 120 -rwxr-xr-x 1 chohjingyi chohjingyi 0 Apr 17 04:20 argument -rw-r--r-- 1 chohjingyi chohjingyi 499 Apr 17 04:20 argument.c -rwxr-xr-x 1 chohjingyi chohjingyi 15968 Apr 17 04:04 hello_world -rw-r--r-- 1 chohjingyi chohjingyi 72 Apr 17 04:04 hello_world.c -rw-r--r-- 1 chohjingyi chohjingyi 84 Apr 17 04:07 hello_world.cpp -rwxr-xr-x 1 chohjingyi chohjingyi 16504 Apr 17 04:09 hello_world2 -rw-r--r-- 1 chohjingyi chohjingyi 551 Apr 17 04:33 input_keyboard.c -rwxr-xr-x 1 chohjingyi chohjingyi 16144 Apr 17 04:37 kernel_info -rw-r--r-- 1 chohjingyi chohjingyi 573 Apr 17 04:37 kernel_info.c -rwxr-xr-x 1 chohjingyi chohjingyi 16192 Apr 17 04:35 keyboard -rw-r--r-- 1 chohjingyi chohjingyi 551 Apr 17 04:35 keyboard.c -rwxr-xr-x 1 chohjingyi chohjingyi 16008 Apr 17 04:36 multiply -rw-r--r-- 1 chohjingyi chohjingyi 545 Apr 17 04:36 multiply.c -rw-r--r-- 1 chohjingyi chohjingyi 545 Apr 17 04:30 multiply.c drwxr-xr-x 3 chohjingyi chohjingyi 4096 Apr 17 04:32 workspace /home/chohjingyi/workspace/workspace: total 8 -rw-r--r-- 1 chohjingyi chohjingyi 551 Apr 17 04:31 input_keyboard.c drwxr-xr-x 2 chohjingyi chohjingyi 4096 Apr 17 04:32 workspace /home/chohjingyi/workspace/workspace/workspace: total 0
	Total Mark:	

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

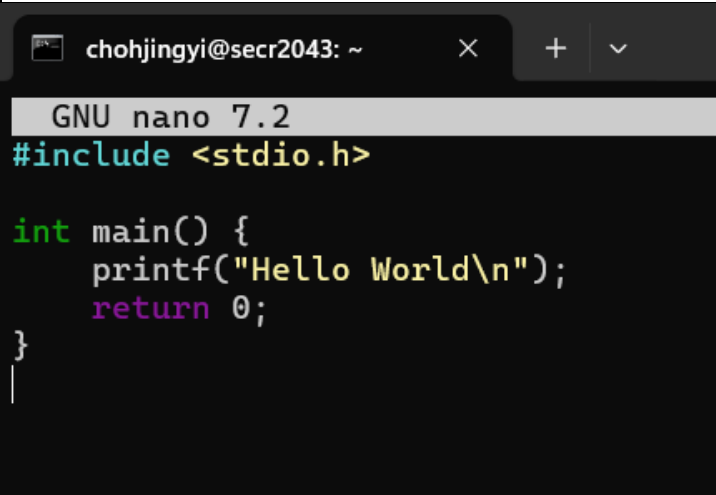
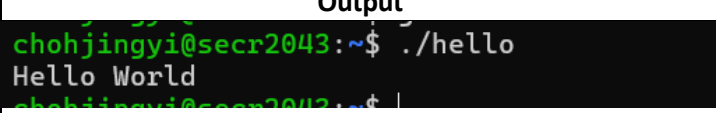
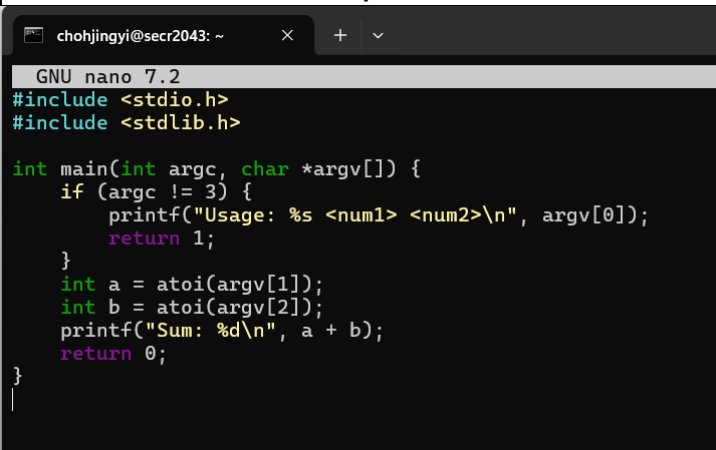
For each question, write down the command and the output (if any) in the answer sheet.

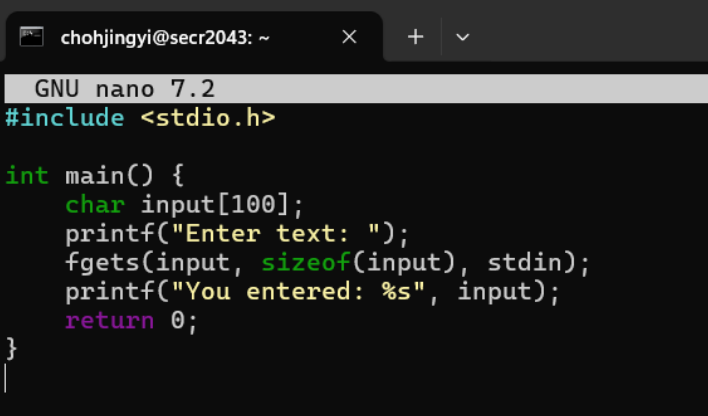
1.	Create an empty file named "hello.txt" in the current working directory.	
	Command	Output
	touch hello.txt	No output
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output
	echo "Hello World" > hello.txt	No output
3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
	echo "This is a test" >> hello.txt	No output
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
	cat hello.txt	chohjingyi@secr2043:~\$ cat hello.txt Hello World This is a test
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
	echo "Goodbye World" > bye.txt	No output
6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
	cat hello.txt bye.txt	chohjingyi@secr2043:~\$ cat hello.txt bye.txt Hello World This is a test Goodbye World
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	
	Command	Output
	cat hello.txt bye.txt > greeting.txt	No output

8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	<pre> chohjingyi@secr2043:~\$ cat greeting.txt Hello World This is a test Goodbye World </pre>
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
	nano greeting.txt	
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	<pre> chohjingyi@secr2043:~\$ cat greeting.txt Hello Choh Jing Yi This is a test Goodbye Choh Jing Yi </pre>
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc. For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
	nano hello.c	
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello".	
	Command	Output
	gcc hello.c -o hello	No Output
3.	Run the executable file "hello" and display its output.	
	Command	Output
	./ hello	
4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".	
	Command	Output
	nano sum.c	

5.	Compile the program "sum.c" using gcc and generate an executable file named "sum".	
	Command	Output
	gcc sum.c -o sum	No Output
6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
	./sum 10 20	<pre> chohjingyi@secr2043:~\$./sum 10 20 Sum: 30 </pre>
7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	
	Command	Output
	./sum -5 15	<pre> chohjingyi@secr2043:~\$./sum -5 15 Sum: 10 </pre>
8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
	nano echo.c	 <pre> chohjingyi@secr2043: ~ GNU nano 7.2 #include <stdio.h> int main() { char input[100]; printf("Enter text: "); fgets(input, sizeof(input), stdin); printf("You entered: %s", input); return 0; } </pre>
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo".	
	Command	Output
	gcc echo.c -o echo	No Output
10.	Run the executable file "echo", type "Hello World" as input and display its output.	
	Command	Output
	./echo	<pre> chohjingyi@secr2043:~\$./echo Enter text: Hello World You entered: Hello World </pre>
	Total Mark:	